



Videofluoroscopy Swallow Study Procedure

1. Scope

Site	Operational Area	Applicable to
Armada Health Service	Speech Pathology Department	Speech Pathology Radiology

2. Purpose

This procedure is aimed to guide Speech Pathology (SP) staff in conducting Videofluoroscopy Swallow Studies (VFSS) in a safe, timely and accurate manner. VFSS is often referred to as one of the 'gold standard' assessments of oropharyngeal dysphagia (Belafsky & Kuhn, 2014b).

The aim of VFSS is:

- To evaluate biomechanical and physiologic function and dysfunction of the oral, pharyngeal and oesophageal swallow;
- Determine swallow safety and efficiency;
- Identify effects of compensatory strategies;
- Determine the appropriate diet and fluid recommendations; and
- Assist in the planning of dysphagia rehabilitation

At AHS, VFSS services are offered to adult patients only.

3. Procedure

3.1 Patient eligibility criteria for VFSS

- Patient must be medically stable enough to leave the ward i.e. oxygen requirements do not exceed what can be provided by portable oxygen
- VFSS will not be performed on patients who are unable to be transferred out of bed
- Patient must be able to cooperate with the assessment tasks and be adequately positioned for the assessment either in their own wheelchair or in the VFSS chair
- Patients whose management outcomes are unlikely to change are not eligible for VFSS. This includes cases of advanced health directives or end of life situation, some chronic conditions or when dysphagia symptoms have resolved.
- All patients must have medical practitioner approval for the procedure (provided through Radiological request form) ([Appendix 1](#))
- Treating SP to gain verbal consent for procedure and document this in medical records. For patients managed by Community/Private SP, the reporting clinician will document consent in either the medical record entry or report.
- Patient must not be allergic to barium or Omnipaque

3.2 Referral Process: Inpatients

- The treating SP to consult with a VFSS competent SP to determine if VFSS is indicated. Treating SP documents bedside/chairside assessment results and rationale for VFSS referral in the patient's medical record.
- The treating SP discusses the rationale for the VFSS with the patient and/or carer/NOK, obtains verbal consent and provides the 'Videofluoroscopy Swallow Study' information brochure ([Appendix 2](#)) which details preparation, procedure and risks.
- The treating SP discusses the rationale for VFSS referral with the medical team and requests referral on a Perth Radiology Clinic request form specifying VFSS ([Appendix 1](#)). Referral forms can be faxed to 93912019 or emailed to PRC Armadale Hospital directly (confirm appropriate contact details via phone prior to emailing).
- The treating SP telephones Radiology on ext. 2012 to arrange a time for the VFSS, ensuring two SP staff members can attend (one of which is VFSS competent).
- The treating SP to complete a Pre-VFSS referral form ([Appendix 3](#)) to be provided to the SP conducting the VFSS, to ensure all relevant information is available.
- The treating SP writes the date and time of the study on the patient journey board, on iSoft nursing handover and informs the Nurse Coordinator on duty so handover can be updated and patient transport to radiology arranged.

3.3 Referral Process: Non-admitted Patients

3.3.1 Medical Team, GP or Community Speech Pathology initiated

- All patients attending VFSS must undergo an initial dysphagia assessment by a SP either from AHS or another facility.
- NDIS/group home residents must have had a dysphagia assessment from a private SP who has recommended VFSS, within 2 months of referral.
- Community SPs referring must complete the 'Pre-VFSS Referral Form' ([Appendix 3](#)) and preferably attend the VFSS with the patient.
- Community SP's must ensure the referring medical practitioner (e.g. GP) has completed a Perth Radiological Clinic request form ([Appendix 1](#)) specifying VFSS and provide it with the Pre-VFSS referral form to AHS SP via SpeechPathology.AHS@health.wa.gov.au.
- Coordinator SP or VFSS competent SP reviews the VFSS referral and determines if suitable; if accepted, email to referrals clerk for entering onto eReferrals - AKG.Referrals@health.wa.gov.au. If not suitable, a response clarifying why it is not suitable to the referral source is required.
- Coordinator SP to determine the patient seating requirements prior to the study. If the patient requires their own wheelchair for seated assessment, dimensions of the chair will be required to determine if the wheelchair size is appropriate for AHS VFSS equipment.

3.3.2 AHS Speech Pathology Initiated Referral

- The SP discusses the rationale for VFSS with the patient and/or carer, obtains verbal consent to pursue referral and provides the 'AHS Patient Information Brochure for VFSS' ([Appendix 2](#)). Treating SP consults with VFSS competent SP regarding appropriateness of referral.
- The SP requests a VFSS referral from the patient's GP ([Appendix 1](#)).
- Upon receipt of the PRC referral, the SP telephones Armadale Perth Radiology Clinic on ext. 2012 to arrange an appointment time.
- The SP informs the patient of the appointment.

- The SP gives the referral to reception staff at Radiology prior to or at the time of the appointment.
- The SP advises OP ward clerks to book the VFSS time/date under VFSS clinic code SPP461.

3.4 Safety

- The SP's must wear a lead apron and lead thyroid cover throughout the examination.
- The Radiographer will minimise radiation exposure for the patient during the procedure by following standard Medical Imaging Guidelines (as per PRC policy).
- The SP will minimise radiation exposure for the patient during the procedure by clearly directing the Radiographer when to commence and cease screening.
- A SP who is, or may be, pregnant must remain behind the screen at all times during the examination.
- A patient who is, or may be, pregnant should discuss with their referring medical specialist, and the duty radiologist regarding potential risks of radiation exposure to the foetus.

3.5 Conducting VFSS

3.5.1 Before Procedure

- The SP will confirm patient identification prior to the procedure using 3 core patient identifiers in accordance with NSQHS Standards, Standard 5 & WA Health Patient Identification and Procedure Matching Policy 2024.
- The SP will complete safety and equipment checks including ensuring working oxygen with face masks, wall suction, tubing and Yankeur sucker are available.
- The SP will prepare the diet and fluid consistencies in accordance [\(Appendix 4\)](#).
- The SP will check for food or contrast allergies prior to study commencing. In the event of a known contrast allergy, an alternative will be sourced in conjunction with Radiographers.
- The SPs assist the Radiographer to transfer the patient into the chair safely and position for screening.

3.5.2 Patients who require a wheelchair

- Non admitted patients who require a wheelchair will have the procedure conducted in their own chair (as long as appropriate dimensions). Arm rests may need to be removed. VFSS may be done in the patient's chair however Anterior- Posterior AP view may not be able to be completed.
- Inpatients who require a hoist transfer will be hoisted into a wheelchair on the ward, prior to being transferred to PRC. Patients must be able to maintain safe seating in an appropriate wheelchair for the duration of the procedure.

3.5.3 During Procedure

- The SP is to liaise with the Radiographer to ensure screening frequency is set to the maximum number of frames per second (gold standard = 30 frames per second).
- The SP is responsible for directing the patient and prompting the Radiographer regarding screening times.
- If a large quantity of contrast is aspirated the managing SP is to arrange chest physio from the patients' ward. Outpatients should be taken to ED for monitoring and/ or chest physio.
- Where anatomical abnormalities are suspected, the SP will request the input of the on-duty Radiologist to assist in making a diagnosis. The SP should give the radiologist a brief verbal handover indicating the noted concern. Should a Radiologist not be available, the procedure

will need to be re-scheduled or recording of the study shown to the on-duty Radiologist when available.

3.5.4 After Procedure

- The SP makes an entry in the patient's notes outlining the main findings and recommendations. If a Radiologist has viewed the procedure, the Radiologist's name and diagnosis should also be documented.
- For inpatients, the treating SP provides feedback to relevant medical and nursing staff as soon as possible (must be on the day of the study) and records any diet and fluid changes on Clinical Manager. For outpatients, the SP discusses the findings with the patient/carer either immediately after the study if time or schedules a time with the patient to discuss at a later time.
- A report is completed within 2 weeks from procedure date and a copy is sent to the referral sources (referring medical officer and referring SP) and uploaded to DMR (using Multimedia uploader; will be located under "Investigations -> Other"). A PDF copy is to be sent to Keenan Tatterton-Down (Keenan.Tatterton-Down@perthradclinic.com.au) so it can be uploaded to PACS. A short entry is written on DMR highlighting that the report has been completed. Historic paper reports were filed in the "Other Results" (Brown divider) section of the patients' medical record.

4. Legislative context

The following documents are required to give effect to this procedure [i.e. the documents included are mandatory]:

- [WA Health Consent to Treatment Policy](#)
- [East Metropolitan Health Service \(EMHS\) Consent Policy](#)

5. Supporting information

The following documents support the implementation of this procedure:

- [SPA VFSS Clinical Guideline SPA VFSS Clinical Guideline.pdf](#)
- Royal Australia and New Zealand College of Radiologists ['Diagnostic Radiology and Pregnancy' position statement](#)

6. Compliance, monitoring and evaluation

The Coordinator of Speech Pathology will be responsible for the compliance of this document. Clinical Incident Management Systems and local speech pathology documentation audit processes will be used to review trends and investigate incidents on an as needs basis/as required.

7. Relevant standards

National Safety and Quality Health Service (NSQHS) Standards (second edition):

- Standard 5 – Action 5.11: Clinicians comprehensively assess the conditions and risks identified through the screening process
- Standard 6 – Action 6.3: Clinicians use organisational processes from the Partnering with Consumers Standard to effectively communicate with patients, carers and families during high-risk situations to:
 - a. Actively involve patients in their own care
 - b. Meet the patient's information needs
 - c. Share decision-making

8. References

- Australian Radiation Protection and Nuclear Safety Agency, 2008. Radiation Protection in Diagnostic and interventional radiology. Retrieved from Safety Guide for Radiation Protection in Diagnostic and Interventional Radiology (arpana.gov.au)
- Belafsky, P. C., & Kuhn, M. A. (2014b). The videofluoroscopic swallow study technique and protocol. In P. C. Belafsky & M. A. Kuhn (Eds.), *The clinician's guide to swallowing fluoroscopy* (pp. 7-13). Springer. https://doi.org/10.1007/978-1-4939-1109-7_2
- Logemann, J.A. (1993). *Manual for the Videofluorographic Study of Swallowing*. Texas: Pro-Ed.
- Royal College of Speech and Language Therapists (2013). Videofluoroscopic evaluation of oropharyngeal swallowing function (VFSS): the role of speech and language therapist. RCSLT Position Paper 2013. London: RCSLT, 2013.
- Speech Pathology Association of Australia (2024). Videofluoroscopy Swallowing Study (VFSS) Clinical Guidelines. Available from: Clinical Guidelines (speechpathologyaustralia.org.au)
- Swigert, N.B (2007), *The Source For Dysphagia*, California: Linguisticsystems

9. Glossary

The following definitions are relevant to this policy.

Term	Definition
Videofluoroscopy	A radiographic instrumental assessment of swallowing
Dysphagia	Difficulty swallowing
Barium	Barium is a chemical element used in the form of Barium meal in the form of an x-ray contrast medium
Omnipaque	Omnipaque is a radiographic contrast medium used for medical imaging.

10. Policy owner (relating to these procedures)

Coordinator Speech Pathology

Enquiries relating to this procedure may be directed to:

Title: A/Coordinator Speech Pathology

Department: Speech Pathology

Email: ashleigh.vidovich@health.wa.gov.au or Rehana.rees@health.wa.gov.au

11. Review

This procedure will be reviewed and evaluated as required to ensure relevance and currency. At a minimum it will be reviewed within one (1) year after first issue and at least every three (3) years thereafter.

Version	Effective from	Effective to	Amendment(s)
V3	09/12/2024	09/12/2027	Scheduled review – minor amendments

12. Authorisation

This procedure has been authorised and issued by the Director Allied Health.

Authorised by	Danielle Kilmurray – Director Allied Health and Ambulatory Care
Authorisation date	03/12/2024
Published date	09/12/2024

Appendix 1: PRC Request Form

ARMADALE HOSPITAL REFERRAL

PERTH RADIOLOGICAL CLINIC		PATIENTS CONTACT NUMBER	☐ Inpatient ☐ Outpatient ☐ Private COPY TO GENERAL PRACTITIONER DR..... SURGERY..... MOBILE.....
Patient Sticker		FEMALE PATIENTS 12-50 YRS DATE OF LMP	☐ Do not send to my Health Record
RADIOLOGIST CONSULTATION REQUEST		HOSPITAL URGENCY	PRC CODE (OFFICE USE ONLY)
☐ X-RAY	Examination required:	☐ Routine ☐ Urgent ☐ ED RESUS ☐ Mobile	WARD 30 / IH 20 / \$ WARD
☐ CT	Examination required:	☐ Routine ☐ Urgent ☐ ED	WARD 60 / IH 60 / ED
☐ MRI	Examination required: ☐ Prostate / Breast / Aneurysm / Other	☐ Routine	WARD
☐ FLUOROSCOPY / IF	Examination required:	☐ Routine ☐ Urgent out of hours (PH, R/S&N)	WARD 20 / WARD
☐ ULTRASOUND	Examination required: ☐ DOPPLER: ☐ Arterial	☐ Routine ☐ Urgent ☐ ED Urgent Consultant to consultant	WARD 120 / IH 60 / \$ PRC Cons; EMHS;
CLINICAL QUESTION			
		Patient is: ☐ Ambulant ☐ Wheelchair ☐ Trolley ☐ Portable ☐ Falls Risk ☐ Spinal Precautions ☐ Interpreter Required	
REQUESTED BY WARD / CLINIC / DEPT Mobile Contact# for results		MAU	
Dr/Mr Provider No.		RISK FACTORS Diabetic ☐ Yes ☐ No Allergies ☐ Yes ☐ No Pregnant ☐ Yes ☐ No BHCQ..... Date Tested	
Doctor Signature Time Date..... Consultant in charge		RENAL FUNCTIONS Urea.....mmol/L Cr.....µmol/L eGFR..... Date Tested IV Cannula ☐ Yes ☐ No	
PRC OFFICE USE			
MODE OF REFERRAL NOTIFICATION ☐ FAX received @ _ _ _ : _ _ APPOINTMENT TIME BOOKED _ _ _ : _ _ _ MIT ARRIVAL @ _ _ _ : _ _ _		PATIENT NOT READY ☐ XOP ☐ XOT ☐ XOR ☐ XOC ☐ XOEQ NOTES:	

perthraddclinic.com.au

PLEASE TELL US IF YOU HAVE HAD PREVIOUS IMAGING SO DIGITAL IMAGES CAN BE RETRIEVED

Appendix 2: VFSS Patient Information Brochure



Government of Western Australia
East Metropolitan Health Service

Armada Kalamunda Group

Videofluoroscopy Swallow Study Patient Information Brochure

What is a Videofluoroscopy Swallow Study (VFSS)?

A VFSS is a moving x-ray procedure that helps assess how a person swallows. During the study, food and drinks are mixed with either barium or Omnipaque, which allows healthcare professionals to observe the swallowing process in real-time using x-ray imaging. It has an important role in:

- Assessing swallowing problems (dysphagia)
- Indicating a need for modified texture food or thickened drinks
- Assessing when silent aspiration is suspected (food or drink passing into the lungs with no cough reflex)
- Determining the effectiveness of safe swallow strategies

What is barium?

Barium sulphate is an insoluble salt and is almost completely nontoxic. It is a white-chalky substance that can be mixed with food and drink and shows up on x-ray.

What is Omnipaque?

Omnipaque is a contrast agent (iodinated) used in medical imaging. It is a clear, water-soluble solution that enhances visibility during x-ray imaging.

*** Both barium and Omnipaque are safe for most patients, though it's important to notify the Speech Pathologist about any allergies or sensitivities.*

Important Points to Remember:

- Videofluoroscopy is held in the Perth Radiology Clinic, located on the ground floor of Armadale Hospital
- If you know you have an allergy to food, please tell the Speech Pathologist before the procedure starts
- Please tell the Speech Pathologist if you could be pregnant before the examination starts
- Please contact the Speech Pathology Department if you are unable to attend your appointment – 9391 2219 or 9391 2395.

What do I need to prepare?

Please bring your Medicare card and any Concession cards you hold (not necessary if you are an inpatient). No fasting is required. You may be asked to remove your glasses and some jewellery prior to the study.

The Procedure

The assessment team includes a Speech Pathologist and Radiographer (technician). A Radiologist (specialist) may be asked to comment on the procedure. You may sit up in a chair, or if able, can stand in the x-ray machine. Your head and neck will be positioned so they can be viewed on the x-ray monitor.

Varying amounts of food and/or drink will be given to you to swallow. Your ability to manage each consistency will be recorded and different techniques may be trialled to try and improve your swallowing function.

How long will it take?

Approximately 30 minutes.

How much does it cost?

Your VFSS will be free if you are a public hospital inpatient or outpatient; or if you have a Concession or Health Care Card.

If you are a private inpatient the invoice will be billed directly to your insurance provider.

What happens after the procedure?

The Speech Pathologist will tell you what the VFSS showed. Strategies and follow up therapy may be recommended, including:

- Food and drink modifications
- Safe swallowing strategies

What are the risks?

Side effects are rare. Barium may cause constipation or bowel obstruction; and less commonly head ache, nausea, or diarrhoea.

Aspiration – Aspiration of barium sulphate or Omnipaque is not expected to cause severe lung injury due to its relatively non-irritant matter. However, if a large amount of contrast goes into the lungs, it may result in chemical pneumonitis or mechanical obstruction of the airways; or in rare instances can be fatal.

If the aspiration of contrast occurs, the Speech Pathologist will contact a hospital based Physiotherapist to assist, or request you visit the Emergency Department for medical clearance. The role of the physiotherapist will be to promote clearance of contrast from the lungs.

Appendix 3: Pre-Videofluoroscopy Referral



Government of **Western Australia**
Department of **Health**
Armadale Health Service

Speech Pathology

Pre-Videofluoroscopy Referral Form

To be completed by the managing speech pathologist

Patient Name:	DOB:
Address:	UMRN:
Telephone:	NOK/Contact Person:
Relationship to Patient:	NOK Contact Details:
Referring Doctor:	Best Appointment Time(s):
Referral Source: ☐ Speech Pathology Department ☐ External Agency ☐ Other	

INFORMED CONSENT

Has the VFSS procedure been explained to the patient?

YES/NO

Has person/carer given their verbal consent for VFSS to proceed?

YES/NO

Reason videofluoroscopy requested

Expected results of videofluoroscopy

BACKGROUND DETAILS

Previous Medical History (please note if diabetic)

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Government of **Western Australia**
Department of **Health**
Armadale Health Service

Speech Pathology

Relevant Case History (i.e., co-existing complaints, recurrent chest infections, education reasons)

Results of clinical bedside assessment

Current Diet & Fluid Recommendations

- ☐ L0 Thin fluids ☐ L2 Mildly thick fluids ☐ L3 Moderately thick fluids ☐ L4 Extremely thick fluids
☐ L7 Regular diet ☐ L6 Soft & bite sized diet ☐ L5 Minced & moist diet ☐ L4 Pureed diet
☐ Nil by mouth ☐ PEG feeding ☐ NGT feeding
☐ Oral trials for pleasure (if so please outline specific recommendations)

Food Allergies (please specify)

Ambulation

- ☐ Independent
☐ Assistance (please specify) _____

Transfers

- ☐ Independent
☐ Assistance (please specify) _____

****Please note – screening apparatus has the capacity to fit wheelchairs less than 150cm wide, and less than 38cm from handles to mid-neck region in size only. The patient will require sitting balance for the procedure, however, may be able to remain in their own wheelchair if needed.**

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Pre-VFSS Referral Form Version Date 04/05/2016

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Government of **Western Australia**
Department of **Health**
Armadale Health Service

Speech Pathology

Cognitive State

- ☐ Normal ☐ Confused ☐ Memory Impairments ☐ Anxious
☐ Depressed ☐ Aggressive ☐ Reduced Insight

Communication

- ☐ No Difficulties ☐ Confused ☐ Comprehension difficulties
☐ Expressive difficulties

Communication method patient uses

- ☐ Verbal ☐ Written ☐ Communication book/board ☐ Gestures

* Please email to AHS.SpeechPathology@health.wa.gov.au *

AKG use only

Action from VFSS request

- ☐ Proceed with videofluoroscopy
☐ Request more information / 2nd opinion
☐ Not proceeding with videofluoroscopy
 ▪ Clinical information sufficient for management
 ▪ VFSS is not expected to change management
 ▪ Patient choice not to have modified diet
 ▪ Factors preventing successful / safe VFSS
 ▪ Other _____

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Pre-VFSS Referral Form Version Date 04/05/2016

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Appendix 4: Omnipaque and Barium Recipes

Equipment required:

- X-Opaque-HD
- Non-Ionic Contrast (NIC)
- Electronic scales
- Pre-prepared diet and fluids
- Precise thickener
- Disposable cups
- Disposable gloves
- Utensils (teaspoon, forks)
- Bowl

Omnipaque (Non-Ionic Contrast, NIC) recipe:

- Use a ratio of one-part Omnipaque to one part apple juice i.e.: 1x50mL Omnipaque 350mg/mL solution to 50mL apple juice (100mLs total of solution).
- To prepare thickened fluids, using Precise Thick-N-Instant, add the following to the 100mL solution:
 - Level 2 mildly thick fluids – 1 pump
 - Level 3 moderately thick fluids – 2 pumps
 - Level 4 Extremely thick - 4 pumps

Barium recipe:

- For L5 / L6 / L7 diet:

Consistency	Barium Sulphate (grams)
Level 5 minced & moist diet (mashed fruit)	40g X-Opaque-HD with 100g level 5 minced & moist fruit
Level 6 soft & bite-sized diet (diced fruit)	40g X-Opaque-HD with 100g level 6 soft & bite sized fruit
Level 7 regular diet (biscuit, sandwich or suitable item)	40g X-Opaque-HD mixed to a paste with water.

- For L4 Puree diet:

Fluid (mL)	Contrast
76 mL	24mL Liquibar (60g)
93 mL	31g X-OPAQUE-HD powder

Measurements calculated via <https://steeleswallowinglab.ca/srri/best-practice/barium-recipes/ddsi-barium-calculator/>

X-OPAQUE-HD = 98% w/w

LIQUIBAR = 125% w/v